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United States Patent	12388008
Kind Code	B2
Date of Patent	August 12, 2025
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Semiconductor interconnect structure with bottom self-aligned via landing

Abstract

A semiconductor structure and method for forming a semiconductor structure includes formation of a recess in a metal layer during the fabrication process to provide process improvements and a conductive via with reduced contact resistance. The semiconductor structure includes a dielectric layer, a metal layer, an etch stop layer, and a conductive via. The top surface of the dielectric layer extends above a top surface of the metal layer, and a bottom surface of the conductive via extends below the top surface of the dielectric layer.

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Appl. No.: 17/376940

Filed: July 15, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230014156 A1	Jan. 19, 2023

Publication Classification

Int. Cl.: H01L23/522 (20060101); H01L21/768 (20060101)

U.S. Cl.:

CPC **H01L23/5226** (20130101); **H01L21/76807** (20130101); **H01L21/76831** (20130101);
 H01L21/76832 (20130101);

Field of Classification Search

CPC: H01L (23/5226); H01L (23/528); H01L (21/76807); H01L (21/76832); H01L (21/76831);
 H01L (24/19)

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Background/Summary

BACKGROUND

(1) Semiconductor devices are used in a wide variety of electronics, and improvements regarding both production and performance of semiconductor devices are generally desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a cross section of an example semiconductor structure, in accordance with some embodiments.

(3) FIG. 2A is a flow chart illustrating an example process for fabricating the semiconductor structure of FIG. 1, in accordance with some embodiments.

(4) FIGS. 2B-2J illustrate cross section of the semiconductor structure of FIG. 1 at each step in the process of FIG. 2A, in accordance with some embodiments.

(5) FIG. 3 illustrates a cross section of a transistor device have contacts connecting to the semiconductor structure of FIG. 1.

(6) FIG. 4 illustrates a cross section of the semiconductor structure illustrating the formation of a conducting via contacting a second metal layer.

DETAILED DESCRIPTION

(7) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(8) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(9) The present disclosure provides a semiconductor structure and method of fabricating a semiconductor structure includes formation of a recess in a metal layer during the fabrication process to provide process improvements and a conductive via with reduced contact resistance. The semiconductor structure includes a dielectric layer, a metal layer, an etch stop layer, and a conductive via. The top surface of the dielectric layer extends above a top surface of the metal layer, and a bottom surface of the conductive via extends below the top surface of the dielectric layer. The etch stop layer can be formed using a thin layer of aluminum oxide and can be partially removed using a wet etching process during the fabrication process. The techniques discussed in the present disclosure can be used to reduce variations in the widths of conductive vias and to enlarge the widths of conductive vias, as well as to control overlap. Further, the techniques discussed herein can be used to provide reduced contact resistance associated with conductive vias as a result of improved landing control of conductive vias, as well as to control metal leak space. The techniques discussed herein can be especially useful in applications with smaller semiconductor devices, such as applications where the pitch is less than 20 nanometers.

(10) Referring now to FIG. 1, a cross section of an example semiconductor structure **100** is shown, in accordance with some embodiments. Semiconductor structure **100** is generally an interconnect structure that provides electrical connections between individual semiconductor devices (e.g. transistors, capacitors, etc.) and conductive metal layers in an integrated circuit. Semiconductor structure **100** can generally provide improvements in terms of reduced contact resistance and improved via landing, as discussed in more detail below. Semiconductor structure **100** can provide improvements in terms of both yield and performance, and can be implemented in a variety of devices and circuits including both memory and processing devices and circuits.

(11) Semiconductor structure **100** is shown to include a first dielectric layer **110**. As shown, the first dielectric layer **110** generally includes a plurality of vertical portions spaced apart from each other. The first dielectric layer **110** can be formed using a variety of suitable deposition processes, including physical vapor deposition, chemical vapor deposition, electrochemical deposition, atomic layer deposition, and other suitable processes and combinations thereof. The first dielectric layer **110** can be formed using a variety of suitable materials, including silicon dioxide, silicon oxynitride, silicon nitride, silicon carbon nitride, silicon oxycarbonitride, hafnia, tantalum nitride,

and other suitable materials and combinations thereof. The first dielectric layer **110** generally provides electrical isolation between different layers of semiconductor structure **100**, as discussed in detail herein.

(12) Semiconductor structure **100** is also shown to include a first metal layer **120**. As shown, the first metal layer **120** is formed adjacent the first dielectric layer **110**. The first metal layer **120** can serve as the first metal layer in an interconnect structure of an integrated circuit, and can thereby be electrically coupled to one or more contacts **190** of one or more semiconductor devices, such as contacts formed on gate, source, and/or drain terminals of individual transistors. The first metal layer **120** can be formed using a variety of suitable deposition and filing processes, including physical vapor deposition, chemical vapor deposition, electrochemical deposition, atomic layer deposition, and other suitable processes and combinations thereof. The first metal layer **120** can be formed using a variety of suitable materials, including ruthenium, tungsten, cobalt, copper, and other suitable materials and combinations thereof. As illustrated in FIG. 1, a top surface **112** of the first dielectric layer **110** extends above a top surface **122** of the first metal layer **120**. In some embodiments, the top surface **112** of the first dielectric layer **110** extends above the top surface **122** of the first metal layer **120** by a magnitude ranging from 2 nanometers to 10 nanometers, however magnitudes outside of this range are also contemplated. The extension of the top surface **112** of the first dielectric layer **110** above the top surface **122** of the first metal layer **120** in this manner can provide a variety of advantages, as discussed herein.

(13) Semiconductor structure **100** is also shown to include an etch stop layer **130**. As shown, etch stop layer **130** is formed adjacent the first dielectric layer **110**, the first metal layer **120**, a second dielectric layer **140**, a conductive via **152**, and a conductive via **154** (discussed in more detail below). In some embodiments, etch stop layer **130** is formed using aluminum oxide, however other suitable materials including silicon nitride, silicon carbide, silicon carbonitride, and various combinations thereof can also be used to form etch stop layer **130**. Etch stop layer **130** can be formed using a variety of suitable deposition processes, including physical vapor deposition, chemical vapor deposition, electrochemical deposition, atomic layer deposition, and other suitable processes and combinations thereof. Use of aluminum oxide to form etch stop layer **130** can provide high etch selectivity and conformity especially when etch stop layer **130** is a thin layer. In some embodiments, etch stop layer **130** is rather thin as a thickness of etch stop layer **130** ranges from 1 nanometer to 15 nanometers, however thicknesses outside of this range are also contemplated. The high etch selectivity of etch stop layer **130** can facilitate improvements in the formation of various other layers within semiconductor structure **100**, such as the second dielectric layer **140** and thereby conductive via **152** and conductive via **154** (discussed in detail below). As illustrated in FIG. 1, portions of etch stop layer **130** are removed in strategic areas such the conductive via **152** and conductive via **154** can be appropriately landed on the first metal layer **120**.

(14) Semiconductor structure **100** is also shown to include a second dielectric layer **140**. As shown, the second dielectric layer **140** generally includes a plurality of different portions spaced apart from each other and strategically formed within semiconductor structure **100**. The second dielectric layer **140** can be formed using a variety of suitable deposition processes, including physical vapor deposition, chemical vapor deposition, electrochemical deposition, atomic layer deposition, and other suitable processes and combinations thereof. The second dielectric layer **140** can be formed using a variety of suitable materials, including silicon dioxide, silicon oxynitride, silicon nitride, silicon carbon nitride, silicon oxycarbonitride, hafnium oxide, tantalum nitride, and other suitable materials and combinations thereof. The second dielectric layer **140** generally provides electrical isolation between different layers of semiconductor structure **100**, as discussed in detail herein.

(15) Semiconductor structure **100** is also shown to include a conductive via **152** and a conductive via **154**. As shown, conductive via **152** and conductive via **154** are formed adjacent the first dielectric layer **110**, the first metal layer **120**, etch stop layer **130**, the second dielectric layer **140**, and a second metal layer **160** (discussed in more detail below). Conductive via **152** and conductive

via **154** generally provide electrical connections between the first metal layer **120** and the second metal layer **160**. Conductive via **152** and conductive via **154** can be formed using a variety of suitable deposition and filing processes, including physical vapor deposition, chemical vapor deposition, electrochemical deposition, atomic layer deposition, and other suitable processes and combinations thereof. Conductive via **152** and conductive via **154** can be formed using a variety of suitable materials, including ruthenium, tungsten, cobalt, copper, and other suitable materials and combinations thereof. As illustrated in FIG. 1, a bottom surface **153** of conductive via **152** and a bottom surface **155** of conductive via **154**, respectfully, extends below the top surface **112** of the first dielectric layer **110**. The extension of the bottom surface **153** of conductive via **152** and the bottom surface **155** of conductive via **154**, respectfully, below the top surface **112** of the first dielectric layer **110** in this manner can provide a variety of advantages, as discussed herein.

(16) Semiconductor structure **100** is also shown to include a second metal layer **160**. As shown, the second metal layer **160** is formed adjacent the second dielectric layer **140**, conductive via **152**, and conductive via **154**. The second metal layer **160** can serve as the second metal layer in an interconnect structure of an integrated circuit, and can thereby be electrically coupled to the first metal layer **120** through conductive via **152** and conductive via **154**. The second metal layer **160** can be formed using a variety of suitable deposition and filing processes, including physical vapor deposition, chemical vapor deposition, electrochemical deposition, atomic layer deposition, and other suitable processes and combinations thereof. The second metal layer **160** can be formed using a variety of suitable materials, including ruthenium, tungsten, cobalt, copper, and other suitable materials and combinations thereof.

(17) Semiconductor structure **100** is also shown to include an insulating layer **170**. Insulating layer **170** is formed adjacent the first dielectric layer **110**. Insulating layer **170** can be implemented as an etch stop layer similar to etch stop layer **130**, for example. Insulating layer **170** can be formed using a variety of suitable materials including silicon nitride, silicon carbide, silicon carbonitride, and various combinations thereof. Insulating layer **170** generally has a higher etch selectivity than the first dielectric layer **110** and can be used to protect different layers such as the first dielectric layer **110** and the first metal layer **120**.

(18) Referring now to FIG. 2A, a flow diagram illustrating an example process **200** for fabricating semiconductor structure **100** is shown, in accordance with some embodiments. FIGS. 2B-2J illustrate cross section of semiconductor structure **100** at each step in process **200**. Process **200** generally includes formation of a recess in metal layer **120**, and forming etch stop layer **130** within the recess to create a bottom self-aligned via landing. Process **200** can be used to reduce variations in the widths of conductive via **152** and conductive via **154** and to enlarge the widths of conductive via **152** and conductive via **154**, and to control overlap. Further, process **200** can be used to provide reduced contact resistance associated with conductive via **152** and conductive via **154** as a result of improved landing of conductive via **152** and conductive via **154** on metal layer **120**. Process **200** can also be used control metal leak space associated with semiconductor structure **100**. Process **200** can be especially useful in applications with smaller semiconductor devices, such as applications where the pitch is less than 20 nanometers.

(19) At a step **201**, a first metal layer is formed adjacent a first dielectric layer (FIG. 2B) and electrically contacts the contacts **190**. As illustrated in FIG. 2B, the first metal layer **120** is formed adjacent the first dielectric layer **110**. In step **210**, the first metal layer **120** can be formed using a variety of suitable deposition and filing processes, including physical vapor deposition, chemical vapor deposition, electrochemical deposition, atomic layer deposition, and other suitable processes and combinations thereof. The first metal layer can be formed using a variety of suitable materials, including ruthenium, tungsten, cobalt, copper, and other suitable materials and combinations thereof. As illustrated in FIG. 2B, the first metal layer is generally formed between vertical portions of the first dielectric layer **110**. The first metal layer **120** can serve as the first metal layer in an interconnect structure of an integrated circuit, and can thereby be electrically coupled to one or

more contacts **190** of one or more semiconductor devices, such as contacts formed on gate, source, and/or drain terminals of individual transistors.

(20) At a step **202**, a first portion of the first metal layer is removed (FIG. 2C). As illustrated in FIG. 2C, a first portion of the first metal layer **120** is removed. In some embodiments, the first portion of the first metal layer **120** can be removed using a chemical mechanical planarization (CMP) process, however other suitable processes and combinations thereof can be used to remove the first portion of the first metal layer **120** in step **202**. The use of a OH process in step **202** can be used to make the top surface **112** of the first dielectric layer **110** even with the top surface **122** of the first metal layer **120**. This smooth surface can facilitate improvements in the formation of semiconductor structure **100**.

(21) At a step **203**, a second portion of the first metal layer is removed (FIG. 2D). As illustrated in FIG. 2D, a second portion of the first metal layer **120** is removed. The second portion of the first metal layer **120** can be removed in step **203** using a variety of suitable processes, including both wet etching and dry etching processes. Wet etchants include, for example, HNO₃, NH₄OH, HCl, HF. Dry etchants include, for example, Cl₂, SF₆, and CF₄. Further, a variety of suitable etchants can be used in step **203**. As illustrated in FIG. 2D, the removal of the second portion of the first metal layer **120** results in a formation of a recess **210** in the first metal layer **120**. After step **203**, the top surface **112** of the first dielectric layer **110** extends above the top surface **122** of the first metal layer **120**. The formation of recess **210** in step **203** ultimately allows for the top surface **112** of the first dielectric layer **110** to extend above the top surface **122** of the first metal layer **120** in the final semiconductor structure **100**. In some embodiments, the depth of recess **210** ranges from 2 nanometers to 10 nanometers, however depths outside of this range are also contemplated.

(22) At a step **204**, an etch stop layer is formed over the first dielectric layer and the first metal layer (FIG. 2E). As illustrated in FIG. 2E, etch stop layer **130** is formed over the first dielectric layer **110**, over the first dielectric layer **120**, and within recess **210**. Etch stop layer **130** can be formed in step **204** using a variety of suitable processes, including physical vapor deposition, chemical vapor deposition, electrochemical deposition, atomic layer deposition, and other suitable processes and combinations thereof. In some embodiments, using atomic layer deposition to form etch stop layer **130** provides advantages in the fabrication process. In some embodiments, etch stop layer **130** is formed in step **204** using aluminum oxide, however other suitable materials including silicon nitride, silicon carbide, silicon carbonitride, and various combinations thereof can also be used to form etch stop layer **130**. Use of aluminum oxide to form etch stop layer **130** can provide high etch selectivity and conformity especially when etch stop layer **130** is a thin layer. In some embodiments, etch stop layer **130** has a thickness that ranges from 1 nanometer to 15 nanometers, however thicknesses outside of this range are also contemplated. As illustrated in FIG. 2E, etch stop layer **130** is also formed within recess **210**.

(23) At a step **205**, a second dielectric layer is formed over the etch stop layer (FIG. 2F). As illustrated in FIG. 2F, the second dielectric layer **140** is formed over etch stop layer **130**. The second dielectric layer **140** can be formed in step **205** using a variety of suitable processes, including physical vapor deposition, chemical vapor deposition, electrochemical deposition, atomic layer deposition, and other suitable processes and combinations thereof. The second dielectric layer **140** can be formed in step **205** using a variety of suitable materials, including silicon dioxide, silicon oxynitride, silicon nitride, silicon carbon nitride, silicon oxycarbonitride, hafnia, tantalum nitride, and other suitable materials and combinations thereof. The second dielectric layer **140** generally provides electrical isolation between different layers of semiconductor structure **100**.

(24) At a step **206**, a portion of the second dielectric layer is removed (FIG. 2G). As illustrated in FIG. 2G, a portion of the second dielectric layer **140** is removed. In some embodiments, the portion of the second dielectric layer **140** is removed in step **206** using a dual damascene process, however other suitable processes including single damascene and other processes can be used to remove the

portion of the second dielectric layer **140** in step **206**. The use of a dual damascene process in step **206** can provide improvements in terms of forming a landing area for conductive via **152** and conductive via **154**. As shown in FIG. **2G**, as a result of step **206**, a recess **222** (for conductive via **152**), a recess **224** (for conductive via **154**), a recess **226** (for a conductive via **156** (not shown in the cross section of FIG. **1**)), and a recess **230** (for the second metal layer **160**) are formed within the second dielectric layer **140**. In some embodiments, additional vias are formed such that vias contact each portion of the first metal layer **120** disposed between the first dielectric layer **110**. In some embodiments, the angle of the sidewalls of the first metal layer **120** are between 85 degrees and 89 degrees, however angles outside of this range are also contemplated. In some embodiments, the angle of the sidewalls of conductive via **152** and conductive via **154** are between 40 degrees and 80 degrees, however angles outside of this range are also contemplated. In some of the embodiments, upper sidewalls of the second dielectric layer **140** may be vertical, while lower sidewalls may be slanted.

(25) At a step **207**, a portion of the etch stop layer is removed (FIG. **2H**). As illustrated in FIG. **2H**, a portion of etch stop layer **130** is removed. In some embodiments, the portion of etch stop layer **130** is removed in step **207** using a wet etching process, however other suitable processes including dry etching processes can also be used to remove the portion of etch stop layer **130** in step **207**. The use of a wet etching process in step **207**, especially in embodiments where etch stop layer **130** is formed using aluminum oxide, can provide more clean and precise removal of etch stop layer **130** to provide a larger and more consistent landing areas for conductive via **152**, conductive via **154**, and conductive via **156**. Accordingly, the widths of conductive via **152**, conductive via **154**, and conductive via **156** can be made larger and conductive via **152**, conductive via **154**, and conductive via **156** can have lower associated contact resistances. The angle of the sidewalls of etch stop layer **130** and the second dielectric layer **140** may be equal or different, depending on the specific materials used and the intended application.

(26) At a step **208**, a conductive via and a second metal layer are formed (FIG. **2I**). As illustrated in FIG. **2I**, conductive via **152** and conductive via **154** are formed along with the second metal layer **160** and conductive via **156**. Conductive via **152**, conductive via **154**, conductive via **156** (with corresponding bottom layer **157**), and the second metal layer **160** can be formed in step **208** using a variety of suitable processes including a variety of suitable deposition and filing processes, including physical vapor deposition, chemical vapor deposition, electrochemical deposition, atomic layer deposition, and other suitable processes and combinations thereof. Conductive via **152**, conductive via **154**, conductive via **156**, and the second metal layer **160** can be formed in step **208** using a variety of suitable including ruthenium, tungsten, cobalt, copper, and other suitable materials and combinations thereof.

(27) At a step **209**, a portion of the second metal layer is removed (FIG. **2J**). As illustrated in FIG. **2J**, a portion of the second metal layer **160** is removed. The portion of the second metal layer **160** can be removed in step **209** using a variety of suitable processes including CMP and other suitable processes and combinations thereof. Step **209** generally includes removing excess material from the second metal layer **160** that extends above a top surface of the second dielectric layer **140** such that a top surface of the second metal layer **160** is even with the top surface of the second dielectric layer **140**. After step **209**, the fabrication process continues with formation of additional insulating and metal layers of the interconnect structure.

(28) The contacts **190** may include, for example, gate contacts, source contacts, and/or drain contacts. FIG. **3** illustrates a transistor device **300** which may include contacts being gate contacts, source contacts, and/or drain contacts. The transistor device **300** of FIG. **3** may be a MOSFET (metal oxide semiconductor field effect transistor). The transistor device **300**, however, is not limited to being a MOSFET, but may be a FinFET, a GAAFET or nanosheet FET, for example. As shown in FIG. **3**, the transistor device **300**, includes a well **310**, gate **312** above the well **310**, source **316A**, drain **316B**, gate oxide **314**, silicide regions **318** and sidewall spacers **320**. In the example

where the substrate **302** is p-type doped, the well **310** may be n-type doped. As such, the transistor device **300** may be an n-type transistor. Further where the substrate **302** is n-type doped, the well **310** may be p-type doped. As such, the transistor device **300** may be a p-type transistor.

(29) In some embodiments, the well **310** may be formed by a doping process to provide the appropriate conductivity type. The well **310** may be formed by exposing a region of the substrate **302** to dopant gas. Alternatively, the well **310** may be formed by implanting a region of the substrate **302** with dopant.

(30) The gate **312** may be formed by depositing a gate material followed by patterning the gate material. The gate material may be, for example, polysilicon, or some other conducting material. The gate **312** may be patterned, for example, by a lithographic process, such as photolithography, or by electron beam patterning. For the photolithographic process a photoresist material may be exposed to light through an appropriately patterned mask, and the photoresist material may be developed and patterned. The patterned photoresist may be used as an etch mask to etch and form the gate **312**. Alternatively a hard mask may be patterned, such as by a photolithographic process, and the patterned hard mask may be used as an etch mask to etch and form the gate **312**. The particular etchant used will depend on the material of the gate **312**. The etching process may be anisotropic, for example, such as a dry etch by reactive ion etching (RIE). The gate oxide **314** may be patterned during the etch process of the gate **312**.

(31) The source **316A** and drain **316B** may be formed in the well **310** using the gate **312** and sidewall spacers **320** as a doping mask. For example, the well **310** may be exposed to an appropriate dopant using the gate **312** and sidewall spacers **320** as a doping mask. The source **316A** and drain **316B** may be formed by exposing the well **310** using a gas to provide dopant to the well **310**. Alternatively, the source **316A** and drain **316B** may be formed by implanting the well **310** with dopant to the well **310**. The source **316A** and drain **316B** may be doped n-type if the well is p-type. The source **316A** and drain **316B** may be doped p-type if the well is n-type.

(32) The silicide regions **318** may be formed on the gate **312**, the source **316A** and the drain **316B** to provide good electrical contact to the gate **312**, the source **316A** and the drain **316B**. The silicide regions **318** may be formed by depositing a silicide forming metal on the gate **312**, the source **316A** and the drain **316B** followed by heating. The silicide forming metal may be Ti or Ta, for example. Heating the silicide forming metal on the gate **312**, the source **316A** and the drain **316B** causes the silicide forming metal to react with the gate **312**, the source **316A** and the drain **316B** forming a silicide. After forming the silicide, the metal which has not reacted may be removed by an etch. The particular etchant will depend on the material of the silicide forming metal.

(33) FIG. 4 illustrates a process for forming a dielectric layer **400**, via opening **410** and conducting via **420** on the structure of FIG. 1 to provide electrical contact to the second metal layer **160**. The dielectric layer **400** may include at least one of: silicon oxide, a comparatively low dielectric constant (k value) dielectric material with a k value less than about 4.0, or combinations thereof. In some embodiments, the dielectric layer **400** is formed of a material, including a low-k dielectric material, an extreme low-k dielectric material, a porous low-k dielectric material, or combinations thereof. The term “low-k” is intended to define a dielectric constant of a dielectric material of 3.0 or less. The term “extreme low-k (ELK)” refers to a dielectric constant of 2.5 or less, and preferably between 1.9 and 2.5. The term “porous low-k” refers to a dielectric constant of a dielectric material of 2.0 or less, and preferably 1.5 or less. A wide variety of low-k materials may be employed by some embodiments of the present disclosure such as, for example, spin-on inorganic dielectrics, spin-on organic dielectrics, porous dielectric materials, organic polymer, organic silica glass, FSG (SiOF series material), HSQ (hydrogen silsesquioxane) series material, MSQ (methyl silsesquioxane) series material, or porous organic series material. The conducting via **420** may be formed by forming a via opening **410** in the dielectric layer **400** followed by forming a conducting material in the via opening **410**. The conducting material may be tungsten, for example. The via opening **410** may be formed in the dielectric layer **400** by a lithographic process, such as

photolithography, or by electron beam patterning. For the photolithographic process a photoresist material may be exposed to light through an appropriately patterned mask, and the photoresist material may be developed and patterned. The patterned photoresist may be used as an etch mask to etch and form the via opening **410** in the dielectric layer **400**. The etching process may be anisotropic, for example, such as a dry etch by RIE.

(34) Once the via opening **410** is formed, the conducting material, such as tungsten, may be formed in the via opening **410** to form the conducting via **420**. The conducting material may then be polished, for example, by chemical mechanical polishing (CMP) to remove the conducting material from a top of the dielectric layer **400** to leave the conducting material only in the via opening **410**. In FIG. **4**, the conducting via **420** electrically contacts the second metal layer **160**.

(35) It will be appreciated that semiconductor structure **100** and process **200** are provided as example implementations, and those skilled in the art will understand that various adaptations to both semiconductor structure **100** and process **200** are contemplated within the scope of the present disclosure. For example, the recess formed in step **203** can be formed deeper or shallower, can be formed only in certain areas of the first metal layer **120**, and can be formed of differing geometries depending on the intended application. Further, the techniques discussed herein can be applied in various areas of an integrated circuit or other electronic components more generally. Further, it will be appreciated that various additional materials and layers not explicitly shown in the figures, such as additional insulating layer, etch stop layers, barrier layers, and other layers can be included depending on the intended application.

(36) As described in detail above, the present disclosure provides a semiconductor structure and method of fabricating a semiconductor structure includes formation of a recess in a metal layer during the fabrication process to provide process improvements and a conductive via with reduced contact resistance. The semiconductor structure includes a dielectric layer, a metal layer, an etch stop layer, and a conductive via. The top surface of the dielectric layer extends above a top surface of the metal layer, and a bottom surface of the conductive via extends below the top surface of the dielectric layer. The etch stop layer can be formed using a thin layer of aluminum oxide and can be partially removed using a wet etching process during the fabrication process. The techniques discussed in the present disclosure can be used to reduce variations in the widths of conductive vias and to enlarge the widths of conductive vias, as well as to control overlap. Further, the techniques discussed herein can be used to provide reduced contact resistance associated with conductive vias as a result of improved landing control of conductive vias, as well as to control metal leak space. The techniques discussed herein can be especially useful in applications with smaller semiconductor devices, such as applications where the pitch is less than 20 nanometers.

(37) An implementation of the present disclosure is semiconductor structure. The semiconductor structure includes a dielectric layer, a metal layer formed adjacent the dielectric layer, and an etch stop layer formed on a top surface of the dielectric layer and on a top surface of the metal layer. The top surface of the dielectric layer extends above the top surface of the metal layer.

(38) Another implementation of the present disclosure is a method of fabricating a semiconductor structure. The method includes forming a recess in a metal layer adjacent a dielectric layer, forming an etch stop layer over the metal layer, within the recess, and over the dielectric layer, removing a portion of the etch stop layer formed within the recess, and forming a conductive via adjacent the metal layer and within the recess.

(39) Yet another implementation of the present disclosure is another semiconductor structure. The semiconductor structure includes a dielectric layer, a metal layer formed adjacent the dielectric layer, and a conductive via formed adjacent the metal layer such that a bottom surface of the conductive via is adjacent a top surface of the metal layer. The bottom surface of the conductive via extends below a top surface of the dielectric layer.

(40) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate

that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a first dielectric layer; a metal layer formed adjacent the first dielectric layer; a contact contacting a bottom surface of the metal layer; an etch stop layer formed on a top surface of the first dielectric layer and on a top surface of the metal layer, the etch stop layer extending over an entirety of a top surface of a first portion of the first dielectric layer and only partially extending over a top surface of a second portion of the first dielectric layer, the first and second portions of the first dielectric layer being separated by the metal layer; and a second dielectric layer including a first portion over a top surface of the etch stop layer and a second portion along sidewalls of the first dielectric layer, wherein the second dielectric layer is a single, continuous layer extending from the first portion to the second portion, and wherein the top surface of the first dielectric layer extends above the top surface of the metal layer.
2. The semiconductor structure of claim 1, further comprising a conductive via formed adjacent the first dielectric layer and the metal layer, wherein a bottom surface of the conductive via extends below the top surface of the first dielectric layer.
3. The semiconductor structure of claim 2, further comprising a second metal layer over and electrically connected to the conductive via.
4. The semiconductor structure of claim 1, wherein the top surface of the first dielectric layer extends above the top surface of the metal layer by a magnitude ranging from 2 nanometers to 10 nanometers.
5. The semiconductor structure of claim 1, wherein the etch stop layer includes aluminum oxide.
6. The semiconductor structure of claim 1, wherein a thickness of the etch stop layer ranges from 1 nanometer to 15 nanometers.
7. The semiconductor structure of claim 1, wherein the second dielectric layer is surrounded by a bottom surface and sidewall surfaces of the etch stop layer.
8. The semiconductor structure of claim 1, wherein the first metal layer directly contacts the contact.
9. A semiconductor structure, comprising: a first dielectric layer; a first metal layer formed adjacent the first dielectric layer; a conductive via formed adjacent the first metal layer such that a bottom surface of the conductive via is adjacent a top surface of the first metal layer, a bottom portion of the conductive via directly contacting sidewalls of the first dielectric layer and a top surface of the first metal layer; a second metal layer formed in the conductive via; and a second dielectric layer formed over the first metal layer, wherein a first portion of the second dielectric layer extends along sidewalls of the first dielectric layer and a second portion of the second dielectric layer extends along sidewalls of the second metal layer, the second dielectric layer extending from the first portion to the second portion as a single, continuous layer, wherein a bottom surface of the second metal layer extends below a top surface of the first dielectric layer, and wherein a top surface of the second metal layer extends above the second dielectric layer.
10. The semiconductor structure of claim 9, wherein the top surface of the first dielectric layer extends above the top surface of the first metal layer.
11. The semiconductor structure of claim 9, further comprising an etch stop layer formed on the top surface of the first dielectric layer and the first metal layer, wherein the etch stop layer is formed using aluminum oxide and a thickness of the etch stop layer ranges from 1 nanometer to 15

nanometers.

12. The semiconductor structure of claim 9, wherein the bottom surface of the second metal layer extends below the top surface of the first dielectric layer by a magnitude ranging from 2 nanometers to 10 nanometers.

13. The semiconductor structure of claim 9, further comprising contacts contacting the first metal layer.

14. A semiconductor structure, comprising: a first dielectric layer; a first metal layer formed adjacent the first dielectric layer; a contact contacting a bottom surface of the first metal layer; a second metal layer contacting top regions of the first metal layer; an etch stop layer formed on a top surface of the first dielectric layer and on a top surface of the first metal layer, the etch stop layer including a first portion that fully covers a top surface of a first portion of the first dielectric layer and a second portion that only partially covers a top surface of a second portion of the first dielectric layer, the first and second portions of the first dielectric layer being separated by the first metal layer; and a second dielectric layer over the first metal layer, the second dielectric layer including a first portion over a top surface of the etch stop layer and a second portion below the top surface of the etch stop layer, the second dielectric layer extending from the first portion to the second portion as a single, continuous layer, wherein a bottom surface of the second dielectric layer extends below the top surface of the first dielectric layer, wherein the top surface of the first dielectric layer extends above the top surface of the first metal layer.

15. The semiconductor structure of claim 14, wherein the top surface of the first dielectric layer extends above the top surface of the first metal layer by a magnitude ranging from 2 nanometers to 10 nanometers.

16. The semiconductor structure of claim 14, wherein the etch stop layer includes aluminum oxide.

17. The semiconductor structure of claim 14, wherein a thickness of the etch stop layer ranges from 1 nanometer to 15 nanometers.

18. The semiconductor structure of claim 14, wherein the second portion of the second dielectric layer is surrounded by a bottom surface and sidewall surfaces of the etch stop layer.

19. The semiconductor structure of claim 14, further comprising a conductive via formed adjacent the first metal layer such that a bottom surface of the conductive via directly contacts the first metal layer.

20. The semiconductor structure of claim 19, further comprising a second metal layer formed over and electrically connected to the conductive via.
